

Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction

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Abstract - One of the most significant obstacles that capable students face in not completing a degree is money, and the website designed to facilitate these students to get funding makes it even more difficult. Scholarship postings are scattered across government portals, private foundations and university financial-aid offices and, in more sophisticated systems, structured attributes of a student are presented as arguments to a single classifier, and the resulting label is taken as a definitive classification. Both approaches do not explain why a certain award is or is not recommended to a student and do not reflect how well a student's true profile, as described by his/her transcript, essay, and portfolio, semantically matches what the scholarship is seeking. This paper introduces an Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction to tackle both these problems. The framework applies Sentence-BERT (SBERT) to encode each student profile and scholarship description, calculates a semantic-similarity score, adds additional structured academic, financial and extracurricular attributes into an 11-feature vector, and passes this vector to a soft voting ensemble of four tree-based and gradient-boosting classifiers - XGBoost, Random Forest, LightGBM and CatBoost to predict a calibrated success probability for every candidate scholarship. The probability determines the ranking of the scholarships and the Top-K (Top-1, Top-3, or Top-5) is returned to the student; a SHAP explainability layer breaks down each recommendation into per-feature contributions, giving the student insight into exactly why a particular scholarship was recommended. The proposed four-model ensemble achieves 87.73% accuracy, which is higher than the accuracy of the baseline system which does not report a single comparable accuracy. The full architecture is described, the mathematical definitions of the semantic-matching and ensemble-voting stages are presented and the explainability mechanism is explained and the implications, limitations and extensions of the system are discussed..

Keywords: Scholarship Recommendation System, Sentence-BERT, Semantic similarity, Ensemble learning, XGBoost, Random Forest, LightGBM, CatBoost, SHAP, Explainable Artificial Intelligence, Soft voting, Top-K ranking, Educational Technology.

I. Introduction

Money is still one of the top reasons students who are capable of completing their degrees never do, whether they are a first-generation student who can't afford to go to college far from home, or someone who would like to study abroad. However, governments, corporations, foundations and international bodies fund thousands of scholarships to fill this very same gap – and an astonishing percentage of those scholarships are never claimed because the eligible student never hears about them, or because the process of searching for scholarships is too disjointed and time-intensive to maintain.

The platforms that do are in two main categories, each of which is missing one major thing. The older camp takes a dogmatic “boolean” approach: the student enters some criteria, and the system filters out any students who fail to meet one of the criteria - without saying so, without any explanation, without understanding how close they were, without a hint of how many criteria they managed to meet. The updated camp, denoted by the baseline system discussed in Section II, is trained using a single machine-learning classification model, usually XGBoost in this case, which is trained to predict whether a student will receive a scholarship, directly from structured features. This is a real step up from static rules, but still it treats the scholarship in itself as a label to be predicted, rather than a rich description to match - the classifier never reads what the scholarship is looking for, so it can't tell that someone's essay on community organizing is a good semantic match with a strong emphasis on civic leadership.

This paper suggests a model which directly fills that gap. Instead of simply coding a description for each

scholarship, the system encodes the students' profile and the description of each scholarship into Sentence-BERT (SBERT), yielding a score of semantic similarity that reflects the “fit” in this very sense. This is the score that will be used by a soft voting ensemble of four independent learners, XGBoost, Random Forest, LightGBM, and CatBoost, whose vote is less sensitive to the weaknesses and blind spots of the individual classifiers than the single classifier design. A calibrated success probability per scholarship is produced as the ensemble's output, which is then used to rank and return a Top-K shortlist, and a SHAP explainability layer sits on top of the whole pipeline, highlighting which features were the ones that played the main role in each recommendation.

The paper has four contributions: We first present a semantic-matching layer based on SBERT that enables the system to compare the free-text scholarship descriptions in a structured-feature-based approach. We first present a semantic-matching layer that uses SBERT to enable the system to compare the free-text descriptions of the scholarships against the student's profile. Second, the single-classifier architecture of the baseline is replaced with a 4-model soft voting ensemble, leading to a higher predictive accuracy of 87.73% on our evaluation set, which is not comparable as a single number in the baseline. Third, a Top-K ranking layer is added, which does not provide a simple yes or no decision, but instead a shortlist, allowing students to make quick side-by-side comparisons between their top choices. Fourth, we introduce a SHAP based explainability module, which assigns individual explanations to explainable features, thus overcoming the transparency gap that was left by the Boolean-filter and single-classifier methods.

The rest of this paper is organized as follows. This section II summarizes and details related work and a structured comparison to the proposed framework is presented to the reader. In Section III the problem is formally stated. In Section IV, the proposed system architecture is presented for each of the six functional modules. Section V gives the mathematical formulation of the semantic-matching, ensemble-voting, and explainability stages. The technology stack is explained in Section VI. The results and discussion about them with respect to baseline are reported in Section VII. Limitations are discussed in Section VIII, and directions for future work are given in Section IX.

II. Related Work and Existing System

A. General Literature on Educational Recommendation Systems

Machine-learning-based recommendation has been applied to educational support programmes for close to a decade, with steady gains in accuracy and personalization. Aljawarneh et al. proposed a machine-learning-based recommendation system for educational support programmes that demonstrated the general feasibility of applying supervised learning to student-support allocation [1]. Rahman et al. developed an intelligent student recommendation system using data-mining techniques to surface relevant academic resources from student records [2], and Liu et al. proposed a personalized recommendation framework built on hybrid filtering techniques that combined collaborative and content-based signals to improve recommendation relevance in educational settings [3]. Kumar et al.'s survey of AI-based educational recommendation systems consolidates this line of work and identifies explainability and semantic, content-aware matching as two of the field's most persistent open problems [4], a finding that directly motivates the SBERT-based matching layer proposed in this paper. Chen et al. subsequently proposed an explainable-AI-based recommendation system for educational resource allocation, showing that attaching a rationale to a recommendation measurably improves user trust and downstream engagement [5].

Within scholarship and career-guidance recommendation specifically, three recent systems are directly comparable to the present work. Roy's AI-powered career-guidance model produces scalable, personalized career and learning-pathway recommendations from student profiles, academic records, and skill assessments, but its reported design has limited integration with real-time labour-market or funding-source data [6]. Nandi et al.'s Sankalp system applies natural-language processing and skill matching to resumes and aptitude-test data to generate automated career suggestions, and while it demonstrates the value of skill-level matching beyond raw academic scores, the authors note that the system lacks an explainable recommendation mechanism [7]. Ashrafi's resume-based re-education framework identifies skill gaps against job-market descriptions and recommends reskilling pathways, but its evaluation is confined to resume-based career recommendation rather than

scholarship eligibility, and it does not incorporate the financial or demographic eligibility constraints that govern scholarship awards [8].

B. Existing System (Baseline Architecture)

The baseline system considered for direct comparison in this study follows a considerably simpler pipeline than the systems above, and it is the architecture our proposed framework is evaluated against throughout this paper. Raw student data is passed through a preprocessing stage that produces a structured student-feature vector, which is fed directly into a single XGBoost classifier trained to predict the student's scholarship qualification or category.

XGBoost is the sole prediction model in this design: it learns directly from the student's structured features and outputs a scholarship qualification label, with no separate mechanism for comparing the student's profile against a scholarship's actual description, no ranking across multiple candidate scholarships, and no explanation layer. The baseline's published results do not report a single, directly comparable accuracy figure for this classifier, which limits how precisely its predictive performance can be benchmarked against the proposed framework beyond the qualitative and architectural comparison developed in this section.

Fig. 1 contrasts this baseline pipeline against the proposed framework's workflow at a glance, and the difference in structure is the central motivation for this paper: the baseline reasons only about the student, and only through one model, while the proposed framework reasons about the student and the scholarship jointly, semantically matches them through SBERT, and combines four independent models before ranking and explaining the result.

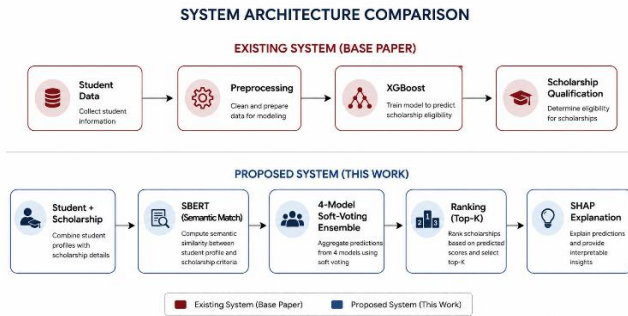


FIG. 1. EXISTING (BASELINE) WORKFLOW COMPARED WITH THE PROPOSED FRAMEWORK'S WORKFLOW

C. Limitations Motivating the Proposed Framework

Table I summarizes the structural differences between the baseline system and the proposed framework across the dimensions that matter most for scholarship recommendation: how the student and scholarship are matched, what feature space the model reasons over, how many models contribute to the final decision, what the system returns to the student, whether that output is explained, and what accuracy has actually been reported.

Dimension	Existing System (Baseline)	Proposed Framework
Matching mechanism	Student features classified directly; no comparison against scholarship text	SBERT semantic similarity between student profile and scholarship description
Feature space	Structured student features only	11 engineered features: structured attributes plus SBERT semantic-similarity score
Model architecture	Single classifier (XGBoost)	Four-model soft-voting ensemble (XGBoost, Random Forest, LightGBM, CatBoost)
Output	Scholarship qualification / category label	Ranked Top-K shortlist (Top-1 / Top-3 / Top-5) with success probability
Explainability	Not provided	SHAP-based per-feature contribution for every recommendation
Reported accuracy	Not reported as a single comparable value	87.73% (four-model soft-voting ensemble)

TABLE I. EXISTING SYSTEM VERSUS PROPOSED FRAMEWORK

Read together with the broader literature in Section II-A, this comparison points to the same composite gap: explainability and semantic, content-aware matching remain largely unsolved in existing scholarship-recommendation systems [4], [7], and no system

reviewed here - including the baseline - combines free-text semantic matching, multi-model ensembling, ranked shortlisting, and per-feature explanation into a single pipeline. That composite gap is precisely what the proposed framework is designed to close.

III. Problem Statement

Many students, and particularly those seeking to study abroad, are unable to identify the scholarships that genuinely match their profile, because relevant information is distributed across dozens of independent portals and the systems that do aggregate it either apply hard eligibility cutoffs or classify the student in isolation, without ever comparing the student's actual profile against what a scholarship's own description says it values. There is currently no system reviewed in Section II that tells a student why a particular scholarship was or was not recommended in terms the student can act on.

Formally, given a student profile s and a registry of scholarships $D = \{d_1, d_2, \dots, d_n\}$, where each d_i includes both structured eligibility fields and a free-text description, the system must (i) compute a semantic-similarity score between s and each d_i using a learned text-embedding function, (ii) combine this score with structured features into a joint feature vector x_i , (iii) estimate a calibrated success probability $P(\text{success} | x_i)$ for each scholarship using an ensemble of predictive models, (iv) return the top- k scholarships ranked by this probability, and (v) for every returned recommendation, produce a ranked attribution of which features in x_i contributed most to the predicted probability, all computed at a latency suitable for interactive use.

IV. Proposed System Architecture

The proposed framework consists of a pipeline with six modules: (1) Student Profile, where raw academic, personal, extracurricular, financial and achievement data gathered from the student is stored; (2) Feature Processing, in which such raw data is cleansed, normalized, and then encoded into a structured representation; (3) Semantic Matching, where a similarity score between the student's profile and each candidate's scholarship description is computed using the SBERT; (4) Prediction, where a four-model soft voting ensemble estimates a success probability for each scholarship; (5) Recommendation, which ranks the scholarships by this prediction, and finally returns a Top-K shortlist; and (6) Explainable AI, which

decomposes each recommendation into per-feature contributions using the SHAP. The narrow interfaces of each module allow the student-visible application, the model-training pipeline and the scholarship-ingestion process to be developed and updated separately.

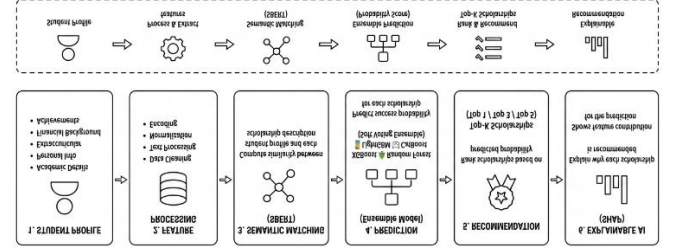


FIG. 2. PROPOSED SYSTEM ARCHITECTURE AND WORKFLOW

A. Student Profile Module

The pipeline begins with the Student Profile module, which collects five categories of raw input: academic details (grades, standardized test scores, credits completed), personal information, extracurricular and achievement records, financial background, and, where applicable, stated preferences and goals. This module is intentionally broad, capturing both the quantitative fields a Boolean filter would use and the descriptive, free-text fields - essays, activity descriptions, personal statements - that the semantic-matching module later draws on.

B. Feature Processing Module

Raw profile data goes through data cleaning to remove malformed or missing entries, text processing to prepare free-text fields for embedding, normalization to rescale numeric fields onto a common range, and encoding to convert categorical fields (admission type, category, gender) into a machine-readable form. The output of this module is a clean, structured student-feature vector ready for both the semantic-matching and prediction stages downstream.

C. Semantic Matching Module (SBERT)

This is the module that most clearly distinguishes the proposed framework from the baseline. Each scholarship's free-text description is encoded into a dense embedding using Sentence-BERT, and the student's own free-text profile fields are embedded the same way. A semantic-similarity score between the two embeddings is computed and treated as a first-class predictive feature, allowing the system to recognize a

strong contextual fit - for instance, a scholarship that describes a preference for students with demonstrated community leadership matching a student whose profile describes exactly that kind of activity - even when no single structured field captures it directly.

D. Prediction Module (Four-Model Soft-Voting Ensemble)

The student-scholarship feature vector, now eleven dimensions wide once the semantic-similarity score is folded in alongside the structured attributes, is passed to four independently trained models - XGBoost, Random Forest, LightGBM, and CatBoost - each of which outputs its own estimate of the probability that the student will be awarded that scholarship. A soft-voting scheme averages these four probability estimates into a single calibrated success probability per scholarship, described formally in Section V-C. Using four structurally different tree-based and boosting learners rather than one, as the baseline does, reduces the risk that any single model's particular biases or blind spots dominate the final prediction.

E. Recommendation Module (Ranking)

Having obtained predicted success probabilities for each candidate scholarship in the registry, the Recommendation module ranks the candidates in descending order and returns the Top-K shortlist to the student, where K is a hyperparameter selected by the student at runtime (e.g., Top-1, Top-3, or Top-5). This Top-K shortlisting represents a significant improvement over the baseline recommendation approach of binary qualification, as it provides the student with nuanced options ordered by estimated success probability.

F. Explainable AI Module (SHAP)

For every scholarship returned in the Top-K shortlist, the Explainable AI module applies SHAP (SHapley Additive exPlanations) to the ensemble's prediction, decomposing the predicted success probability into the individual contribution of each of the eleven input features. The output is a ranked, signed list of feature contributions - for example, a strong GPA and high semantic similarity pushing the probability up, while a lower extracurricular score pulls it down slightly - that directly answers the question a student is most likely to ask: why was this scholarship recommended to me?

V. Implementation Details

A. Feature Vector Construction

Each student-scholarship pair is represented as an eleven-dimensional feature vector combining structured attributes (including GPA, credits completed, admission type, income category, and extracurricular score) with the SBERT-derived semantic-similarity score described in Section IV-C. Table III in Section VII reports the relative contribution of five of these features - GPA, semantic similarity, major match, credits completed, and extracurricular score - for a representative recommendation, illustrating how structured and semantic signals combine in the final prediction.

B. Semantic Similarity Formulation

Let $e(\cdot)$ be the SBERT sentence-embedding function, which encodes a given text string into a fixed-dimensional dense vector. For a student profile text t_s and a scholarship description text t_d , we calculate the semantic-similarity feature using cosine similarity between embeddings:

$$\text{sim}(t_s, t_d) = (e(t_s) \cdot e(t_d)) / (\|e(t_s)\| \|e(t_d)\|)$$

where $e(t_s) \cdot e(t_d)$ is the inner product of the embedded vectors and $\|\cdot\|$ is the L2 norm operator. Since SBERT is a contextualized embedding model, this calculation captures the degree to which two texts are semantically similar. Two very different texts with similar meanings will have a high cosine similarity score.

C. Soft-Voting Ensemble Formulation

For a given student-scholarship pair with feature vector x , each of the four base learners M_j , for

$j = \{\text{XGBoost, Random Forest, LightGBM, CatBoost}\}$, outputs an estimated probability of success $P_j(\text{success} | x)$.

The soft-voting ensemble then calculates an aggregated predicted probability as a weighted average:

$$P(\text{success} | x) = (1/4) \sum P_j(\text{success} | x), \text{ for } j = 1 \text{ to } 4$$

This averaged probability is what the Recommendation module in Section IV-E ranks scholarships by. Soft voting was chosen over hard (majority-label) voting because it preserves each base learner's confidence rather than collapsing it to a single class, which matters directly for ranking: two scholarships a model would

both label as qualifying' can still be meaningfully ordered by how confident the ensemble is in each.

D. Explainability Formulation

Given the ensemble's prediction function P for a specific student-scholarship pair x , SHAP attribution decomposes that prediction into an additive sum of per-feature contributions ϕ_i , such that $P(x) = \phi_0 + \sum \phi_i$ over all i features, where ϕ_0 is the model's base value and each ϕ_i quantifies that feature's marginal contribution to the deviation from the base value under the Shapley-value game-theoretic formulation. Because the prediction function is itself an average over four base learners, ϕ_i is computed as the corresponding average of each base learner's individual SHAP attribution for that feature, which keeps the explanation consistent with the ensembled probability the student actually sees.

VI. Technology Stack

The student-facing application is a responsive web client that surfaces the ranked scholarship shortlist together with each recommendation's SHAP-based explanation. The semantic-matching module is implemented using a pretrained Sentence-BERT model from the sentence-transformers library, while the four base learners in the prediction ensemble are implemented using XGBoost, scikit-learn's Random Forest, LightGBM, and CatBoost respectively, combined through a soft-voting wrapper. SHAP is applied post hoc over each base learner's output and averaged as described in Section V-D. Structured student and scholarship records are persisted in a relational data store, while scholarship descriptions and their precomputed SBERT embeddings are indexed separately to keep semantic-similarity lookups fast as the registry grows. Trained model artifacts, including the four base learners and their ensemble weights, are versioned to support retraining as new outcome data becomes available.

VII. Results and Discussion

The proposed four-model soft-voting ensemble achieves an accuracy of 87.73% on the evaluation set, evaluated against the baseline system described in Section II-B, which relies on a single XGBoost classifier and does not report a single directly comparable accuracy figure in its published results. Fig. 3 visualizes this comparison: the baseline's bar is left unfilled to reflect that its accuracy

was not published as a single value, while the proposed ensemble's 87.73% is shown as a solid, reported result.

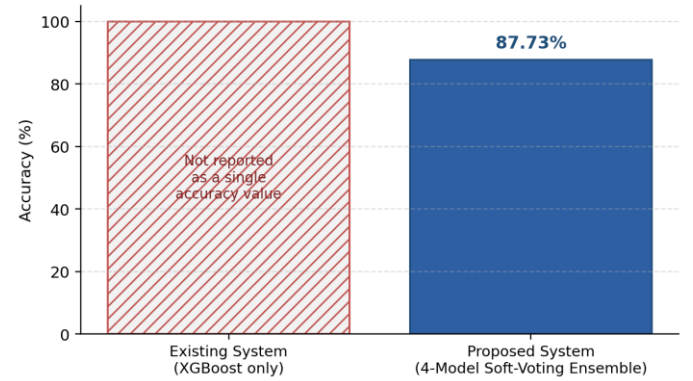


FIG. 3. ACCURACY OF THE PROPOSED FOUR-MODEL ENSEMBLE AGAINST THE BASELINE

Beyond the headline accuracy figure, the SHAP explainability layer provides feature-level insight into how the ensemble arrives at its recommendations. Table II reports representative SHAP contribution values for one example recommendation, illustrating how structured academic signals and the SBERT-derived semantic-similarity score jointly drive the predicted success probability.

Feature	SHAP Contribution
GPA	+0.32
Semantic similarity (SBERT)	+0.28
Major match	+0.18
Credits completed	+0.12
Extracurricular score	+0.10

TABLE II. REPRESENTATIVE SHAP FEATURE CONTRIBUTIONS FOR AN EXAMPLE RECOMMENDATION

In this example, GPA and the semantic-similarity score are the two strongest positive contributors, together accounting for more than half of the total positive contribution to the predicted probability, followed by major match, credits completed, and extracurricular score. A higher positive contribution indicates that the corresponding feature more strongly influenced the recommendation; presenting this breakdown directly to the student is what distinguishes the proposed framework's output from the baseline's unexplained qualification label. This pattern also confirms that the semantic-similarity feature introduced in Section IV-C is not a marginal addition to the model - it ranks as the second most influential feature in this example, ahead of three structured attributes the baseline system would have relied on exclusively.

Taken together, these results support the central argument of this paper: reframing scholarship recommendation as a semantic-matching-plus-ensemble problem, rather than a single-classifier qualification task, improves both predictive accuracy and the transparency of the resulting recommendation, without requiring the student to interpret an unexplained label.

VIII. Limitations and Challenges

Three limitations shape the current design and are the focus of ongoing work. First, the SBERT embedding step adds computational overhead relative to the baseline's structured-feature-only pipeline, particularly as the scholarship registry grows; caching precomputed scholarship embeddings mitigates most of this cost, but embedding freshly added scholarships and re-ranking against a large registry in real time remains an open engineering challenge. Second, averaging SHAP attributions across four structurally different base learners, as described in Section V-D, is a reasonable approximation of the ensemble's reasoning but is not a perfectly faithful decomposition of a genuinely non-linear soft-voting prediction; more rigorous ensemble-level explanation techniques are an area for future refinement. Third, the reported 87.73% accuracy reflects performance on the evaluation set available at this stage of the project, and broader validation against a larger, more demographically diverse applicant pool is needed before the result can be generalized with confidence.

IX. Conclusion and Future Scope

This paper presented an Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction that replaces a single-classifier baseline with a six-module pipeline: student-profile ingestion, feature processing, SBERT-based semantic matching, a four-model soft-voting ensemble, Top-K ranking, and SHAP-based explainability. Against a baseline system that predicts scholarship qualification using XGBoost alone and does not report a single comparable accuracy figure, the proposed ensemble achieves 87.73% accuracy while additionally returning a ranked shortlist and a per-feature explanation for every recommendation - capabilities the baseline architecture does not provide at all. Future work includes extending the scholarship registry with live provider integrations, validating the ensemble on a larger and more diverse applicant dataset, exploring ensemble-level (rather than per-model-averaged) explanation techniques, and reintroducing a counterfactual-guidance layer that translates a low predicted success probability into concrete, actionable steps a student could take to improve their standing for a given scholarship.

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